

DEEP-DIVE ANALYSIS & INTERACTIVE DASHBOARD

Task - 3 Report

Objective

The objective of this project is to perform a comprehensive analysis of the sales dataset by defining Key Performance Indicators (KPIs), conducting a deep-dive business analysis, and developing an interactive dashboard. The dashboard enables users to monitor business performance, explore trends, and make informed decisions using interactive visualizations and filters.

Dataset Description

The project uses a cleaned sales dataset prepared during Task 2 (Exploratory Data Analysis).

The dataset contains information related to:

- Customer Details
- Product Categories
- Sales Amount
- City
- Gender
- Orders
- Monthly Sales

The dataset was cleaned by handling missing values, removing duplicates, and formatting the data appropriately for visualization and analysis.

Core KPIs

1. Total Sales

Formula

SUM(Total Sales)

Purpose

Measures the total revenue generated from all customer orders.

2. Total Orders

Formula

COUNT(Order ID)

Purpose

Represents the total number of customer orders processed.

3. Total Customers

Formula

COUNT(DISTINCT Customer ID)

Purpose

Indicates the number of unique customers who made purchases.

4. Average Order Value

Formula

Average Order Value = Total Sales / Total Orders

Purpose

Shows the average amount spent per order.

5. Revenue Per Customer

Formula

Revenue Per Customer = Total Sales / Total Customers

Purpose

Measures the average revenue generated from each customer.

Deep-Dive Analysis

For this project, **Customer Segmentation and Sales Performance Analysis** were selected as the primary business focus.

The analysis includes:

- Sales by City
- Sales by Product Category
- Monthly Sales Trend
- Gender-wise Sales
- Customer Performance
- Revenue Distribution

These analyses help identify business growth opportunities and customer purchasing behavior.

Dashboard Features

The dashboard was developed using **Python Streamlit** with interactive visualizations created using **Plotly**.

The dashboard provides:

- KPI Cards
- Interactive Filters
- Sales by City
- Sales by Category
- Monthly Sales Trend
- Gender-wise Sales
- Top Customers
- Dynamic Data Filtering

Dashboard Screenshots

Figure 1

Dashboard Home



Figure 2

Interactive Filters

localhost:8501

City

Bengaluru × Kolkata × Hyderabad × Patna × Mumbai × nan ×

Delhi × Gaya × Pune ×

Gender

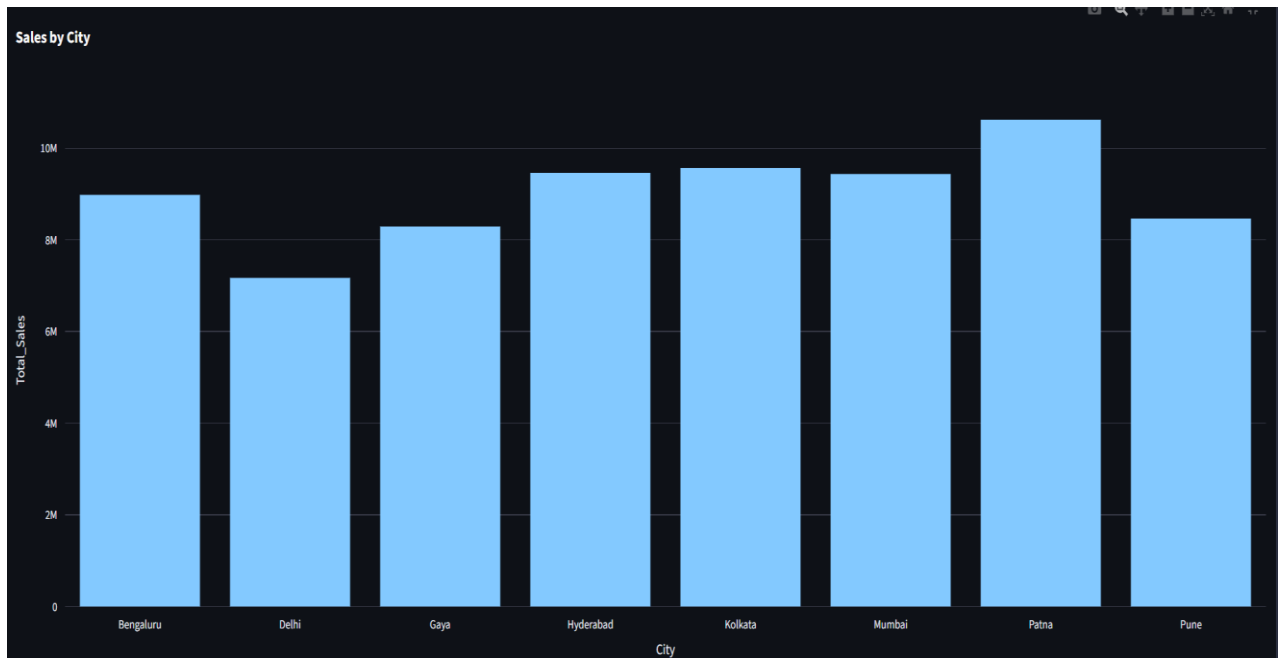
Male ×

Female

Figure 3

Sales Analysis Charts





Key Findings

Based on the dashboard analysis:

- The business generated significant overall sales revenue.
- Electronics contributes the largest share of total sales.
- Sales vary across different cities, indicating regional market differences.
- Customer purchasing behavior differs by gender.
- KPI cards provide a quick overview of overall business performance.
- Interactive filters allow users to analyze data based on selected cities and gender.

Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Increase marketing efforts in high-performing cities.
2. Promote top-selling product categories to maximize revenue.
3. Introduce loyalty programs for frequent customers.
4. Improve marketing campaigns in lower-performing cities.
5. Use dashboard insights for strategic business planning and inventory management.

Technologies Used

- Python

- Pandas
- Streamlit
- Plotly
- VS Code

Project Deliverables

The following files were developed during Task 3:

- Interactive Dashboard (dashboard.py)
- Cleaned Dataset (cleaned_dataset.csv)
- Project Report (Task3_Report.pdf)
- GitHub Repository
- README File

Conclusion

This project successfully demonstrates how business data can be transformed into meaningful insights using interactive dashboards. The defined KPIs and deep-dive analysis provide a clear understanding of sales performance, customer behavior, and revenue trends. The Streamlit dashboard enables users to interact with the data efficiently and supports better business decision-making through real-time visualizations.